

Partner-Specific Affective Precision Gates Social Policy Deployment in Multi-Agent Active Inference

TODO

Draft 1 - manuscript packet

Abstract

Active-inference accounts of affect model valence as an inference over expected action precision: affect reports how well an agent expects its action model to fit the world. Social interaction adds a factorization problem. A focal agent does not merely need a global estimate of whether the world is predictable; it must track which social partners are predictable, volatile, exploitable, or worth probing. We extend Hesp-style affective precision to a repeated multi-partner trust task by assigning each partner a local model-fitness signal. Each partner is represented by a partner-local `pymdp.Agent` over hidden partner type, partner stance, and own-action bookkeeping, while an external beta tracker updates from partner-action prediction error and modulates partner-local policy precision through $\gamma_k = \gamma_{\text{base}}/\mathbb{E}[\beta_k]$. Across current-architecture simulations, partner-local precision is behaviorally active when the policy posterior has room to move, tracks prediction reliability more strongly than realized payoff, and changes policy deployment even when belief readouts remain similar. The strongest positive mechanism result is a 30-seed reliability-versus-reward confirmation in which $|\text{corr}(\text{precision}, \text{surprise})| = 0.701$ and $|\text{corr}(\text{precision}, \text{payoff})| = 0.419$, with no affect payoff advantage. The strongest boundary result is a 30-seed abrupt-betrayal confirmation in which affect lowered policy entropy but reduced whole-run payoff relative to no affect. These results support affective precision as a partner-specific model-fitness and deployment signal, not as a generic reward booster. The simulations do not validate clinical phenotypes or human behavior; they define a mechanistic testbed and a set of boundary conditions for future empirical and model-comparison work.

Draft status. This manuscript is a first full write-up from the current evidence packet. It intentionally uses narrow claims, TODO citation placeholders, and draft figures. It should be updated after additional confirmation runs, global-beta ablation, final figure redraws, and citation verification.

1 Introduction

Active inference treats perception and action as coupled inference under a generative model. An agent maintains beliefs about hidden causes of observations, evaluates policies by expected free energy, and selects actions by a softmax over policy values weighted by policy precision [TODO-CITE: active inference and expected free energy]. In this framework, policy precision is not a decorative temperature parameter. It determines whether the agent acts decisively on the currently preferred policy or preserves a more exploratory distribution over plausible policies.

Affective active-inference accounts give policy precision psychological content. In Hesp et al.’s account of deeply felt affect, affective valence is tied to expected action precision and subjective model fitness: positive affect corresponds to an expectation that the current generative model is working well enough to support confident action, while negative affect corresponds to model failure or reduced expected precision [TODO-CITE: Hesp et al. *Deeply Felt Affect*]. This converts affect from an externally added reward bonus into a metacognitive signal about whether the agent can trust its own model.

Social interaction makes this model-fitness problem naturally multi-factorial. A focal agent interacting with several partners may have a reliable model of one partner, an unreliable model of another, and a confidently adversarial model of a third. A single global affective precision signal collapses these distinctions. In social environments, the relevant question is often not “Do I feel good?” or “Has this partner rewarded me?” but “How well does my model of this partner currently work?” A predictable exploiter can have high model fitness even if interacting with them is unrewarding; a volatile ally can have low model fitness even if their average payoff is favorable.

This paper tests a partner-specific extension of affective precision in a repeated trust task. The central claim is that affective precision gates deployment of otherwise available social knowledge. It is not modeled as a partner liking score, a reward cache, or a literal trust scalar. It is an auxiliary partner-local estimate of the reliability of the focal agent’s own social model. The signal updates from prediction error about the partner’s action and modulates the policy posterior used to choose actions or partners.

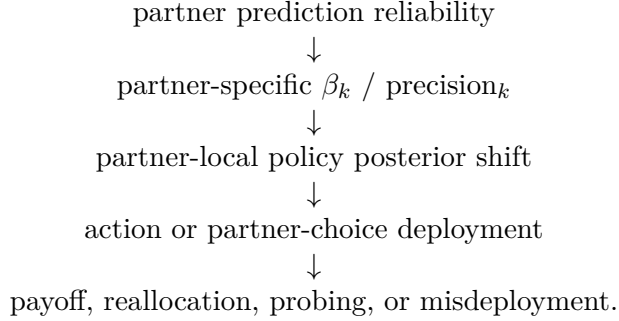
The project contributes four elements. First, it factorizes Hesp-style expected action precision over social partners, making affect a set of partner-local model-fitness estimates rather than a single global action-model state. Second, it implements the mechanism around official `inferactively-pymdp==1.0.0` agents: state and policy inference remain in standard partner-local POMDP agents, while beta tracking is an external task-owned module. Third, it tests the mechanism with readouts that separate model fitness from reward, deployment from belief accuracy, social choice from payoff, and stress amplification from recovery success. Fourth, it reports a productive boundary condition: abrupt betrayal can turn the same precision channel into misdeployment by sharpening action under a wrong or insufficiently recalibrated post-switch model.

The resulting claim is deliberately narrower than “affect improves payoff.” In open graded regimes, partner-specific precision can lower entropy and improve payoff. In reliability-versus-reward tests, it follows predictive reliability more strongly than payoff. In social-choice tests, it changes partner-selection distributions before payoff is informative. Under abrupt betrayal, however, affective precision can lower entropy while worsening payoff. This combination is theoretically useful because it shows that the mechanism is active but not uniformly beneficial.

2 Theory: Affective Precision as Partner-Local Model Fitness

2.1 From global affect to partner-specific affect

The proposed mechanism is:



The affective state is not a hidden state inside the trust POMDP. It is an external precision tracker coupled to the dynamics of inference. For partner k , the tracker reads how surprising the partner’s observed action was under the focal agent’s current partner-local beliefs. Accurate predictions push the tracker toward lower beta and therefore higher expected policy precision; surprising observations push it toward higher beta and therefore lower expected policy precision.

This gives the model a different interpretation from reward averaging. A reliable cooperator and a reliable exploiter can both be predictable, so both can support high model fitness. The action implied by high precision differs: for the cooperator, high precision may support confident approach or cooperation; for the exploiter, it may support confident avoidance or defection. Conversely, a partner who sometimes rewards the agent but changes unpredictably should not necessarily produce high precision because the agent’s model is not stable enough to deploy confidently.

2.2 Deployment rather than knowledge

The model’s deployment claim is inspired by the knowing-versus-using dissociation associated with somatic-marker accounts of decision-making [TODO-CITE: Damasio; Bechara Iowa Gambling Task]. The paper does not implement vmPFC, skin conductance, or a neural circuit. It implements a computational analogue: a lesioned agent can maintain or update the partner-specific beta state, and can retain broadly similar partner beliefs, but the beta-to-policy-precision pathway is decoupled. When the policy posterior is open enough for precision to matter, this lesion should impair action deployment without necessarily destroying social inference itself.

This distinction matters for interpreting negative results. If beliefs are intact but action is worse, affect is not being used as a knowledge source; it is being used as a policy-precision gate. If beliefs and behavior are both unchanged, either the mechanism is not active or the policy space is saturated. If affect lowers entropy and makes behavior worse after a shock, the mechanism is active but miscalibrated for that transition regime.

2.3 Hypothesis surface

The current manuscript uses six behavior cards. H0 asks whether the policy posterior has room to move. H1 tests model fitness against reward. H2 tests deployment by decoupling beta from policy precision. H3 tests stress and betrayal as volatility boundary conditions. H4 tests partner selection. H5 tests clinical-like parameter perturbations as precision-dynamics variants, not as diagnoses.

Table 1: Evidence hierarchy used in this draft. H1 and H3 are the primary statistical anchors because they use 30 seeds per variant. H0, H2, H4, and H5 are supporting or supplemental because they use five seeds per variant.

Card	Mechanism role	Seeds	Draft status
H0	Policy openness gates visible affect effects	5/variant	Supporting
H1	Precision tracks model fitness rather than reward	30/variant	Primary
H2	Affect changes deployment despite similar beliefs	5/variant	Supporting
H3	Stress exposes both deployment and misdeployment	30/variant	Primary
H4	Partner-specific precision guides social choice	5/variant	Supporting
H5	Clinical-like perturbations separate precision dynamics	5/variant	Supplemental

3 Model

3.1 Trust-game architecture

The simulations use a repeated trust task with a focal active-inference agent interacting with multiple social partners. The primary trust task is a scripted-partner surface designed to isolate the focal agent’s inference and deployment mechanism; fully reciprocal active-inference partners are a future extension rather than the evidence base for the present claims.

For each partner, the focal agent maintains one partner-local `pymdp.Agent`. The partner-local POMDP has three hidden-state factors:

$$s_t = (s_{\text{type}}, s_{\text{stance}}, s_{\text{own}}). \quad (1)$$

The type factor has four states: cooperator, reciprocator, exploiter, and random. The stance factor has three states: trusting, neutral, and hostile. Stance is the partner’s disposition toward the focal agent, not the focal agent’s attitude. It is hidden from the focal agent and evolves through action-dependent transition dynamics. The own-action factor is deterministic bookkeeping that records the focal agent’s previous action so that payoff can be represented as a standard observation likelihood.

The model uses two observation modalities. The first is the partner’s action, cooperate or defect, generated from partner type and stance. The second is payoff, generated from own action, type, and stance after marginalizing over the partner action. In the environment, realized payoff is deterministically derived from joint actions. In the generative model, payoff is a likelihood modality over payoff levels, which preserves the incentive structure while staying within standard factorized POMDP conventions.

Partner-local controls are factorized. A singleton partner-control placeholder is used inside each partner-local agent. A stance-control factor drives action-dependent stance transitions during planning, and an own-action factor determines the executed investment/action and payoff likelihood. In agent-choice runs, partner choice is handled outside each partner-local POMDP by evaluating candidate decisions for each partner and routing the final executed action to the selected partner.

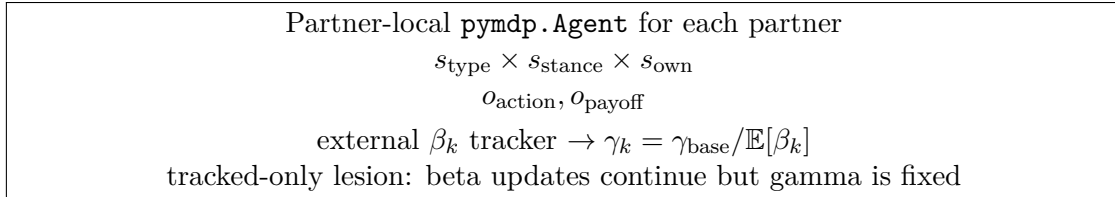


Figure 1: Partner-specific affective precision architecture. Each partner has a local POMDP for social inference. The beta tracker is external to the POMDP and modulates policy precision rather than adding another sensory modality or hidden-state factor.

3.2 Observation and transition structure

The partner-action likelihood maps type and stance to cooperation probability. For example, a cooperator in a trusting stance is highly likely to cooperate, while an exploiter in a hostile stance is unlikely to cooperate. The payoff likelihood uses the standard trust-game payoff table, with payoff levels -1 , 1 , 3 , and 5 corresponding to the outcomes of the focal and partner actions.

The type transition is near-identity with small stochastic drift, allowing nonstationarity without making type arbitrarily volatile. The stance transition is action dependent. Cooperating tends to build trust gradually, moving neutral partners toward trusting and hostile partners slowly toward neutral. Defecting can destroy trust quickly, moving trusting partners toward hostility. This action-dependent stance structure is what makes planning depth meaningful in the current architecture.

Preferences are placed over payoff observations rather than directly over partner actions. The agent has no intrinsic preference for observing cooperation or defection by the partner, except insofar as those observations are informative or connected to payoff. Policy priors are uniform in the supported runs unless otherwise specified.

3.3 External beta tracker

The beta tracker operates after each partner interaction. For partner k , the tracker computes a bounded prediction-error proxy from the probability assigned to the observed partner action before the update:

$$\epsilon_k = 1 - P(o_k^{\text{action}} = o_k^{\text{obs}} \mid q(s_{\text{type}}, s_{\text{stance}})). \quad (2)$$

The tracker converts this unsigned prediction error into signed affective charge:

$$\phi_k = \alpha(\sigma_0^2 - \epsilon_k^2). \quad (3)$$

When squared prediction error is below the baseline surprise scale σ_0^2 , the charge is positive and favors lower beta, which corresponds to higher expected policy precision. When squared prediction error is above baseline, the charge is negative and favors higher beta, corresponding to lower expected policy precision.

The tracker represents $q(\beta_k)$ as a categorical distribution over discrete beta levels. After applying a persistence transition matrix, it forms a pseudo-likelihood from the affective charge and effective precision at each beta level:

$$\text{effective_precision}_\ell = 1/\beta_\ell, \quad (4)$$

$$\log L_\ell(\phi_k) = \phi_k \cdot \text{effective_precision}_\ell, \quad (5)$$

$$q(\beta_k)' \propto L(\phi_k) \odot T_\beta q(\beta_k). \quad (6)$$

The expected beta then modulates the partner-local policy precision:

$$\gamma_k = \frac{\gamma_{\text{base}}}{\mathbb{E}_{q(\beta_k)}[\beta_k]}. \quad (7)$$

Policy selection follows the standard active-inference softmax over expected free energy, with the partner-local precision applied before policy inference:

$$q(\pi | k) = \sigma(-\gamma_k G(\pi)). \quad (8)$$

Low beta therefore means high policy precision and more decisive deployment. High beta means low policy precision and a more diffuse policy posterior.

3.4 Variants and lesions

The affect variant updates beta and uses γ_k during policy selection. The no-affect or decoupled baseline fixes the effective policy precision to γ_{base} . The tracked-only lesion keeps beta updates available but decouples beta from policy selection by forcing $\gamma_k = \gamma_{\text{base}}$. This is the key deployment lesion: the agent can track partner-specific model fitness, but that state cannot sharpen or soften action selection.

The H5 variants perturb beta dynamics. Alexithymia-like variants dampen the charge update so beta moves weakly. Borderline-like variants increase gain or volatility so beta moves too strongly. Depression-like variants bias the initial state or dynamics toward lower expected precision. These labels are computational shorthand only. They do not validate clinical diagnoses, and the manuscript treats them as parameter perturbations of precision dynamics.

4 Experiments and Analysis

4.1 Experiment cards

H0 tests whether policy-space openness gates observable affect effects. If the policy posterior is already saturated, beta can move without changing behavior. Open graded action spaces, partner-choice regimes, and volatility windows should leave more policies alive and therefore expose precision effects.

H1 tests whether partner-local precision tracks predictive reliability rather than realized reward. The critical dissociation is a predictable low-reward partner: a reward cache should be low, while a model-fitness signal can be high if the agent can predict the partner.

Table 2: Audited run sizes included in this draft.

Experiment	Result role	Variants	Seeds	Rounds
H0 graded choice	Supporting openness evidence	2	5 each	200
H2 lesion open	Supporting deployment evidence	3	5 each	200
H4 partner choice	Supporting social-choice evidence	2	5 each	200
H1 confirmation	Primary model-fitness evidence	2	30 each	200
H3 confirmation	Primary betrayal boundary evidence	3	30 each	120
H3 abrupt sensitivity	Primary shock-regime follow-up	12	30 each	120
H3 gradual sensitivity	Primary shock-regime follow-up	8	30 each	120
H5 clinical dynamics	Supplemental perturbation dynamics	4	5 each	200
H5 clinical betrayal	Supplemental perturbation dynamics	4	5 each	120
H5 affect sensitivity	Supplemental perturbation dynamics	6	5 each	120

H2 tests deployment by comparing affective agents with tracked-only or no-affect controls. The expected pattern is that belief readouts remain similar while policy entropy, payoff, or partner choice diverges in open regimes.

H3 tests stress. Abrupt stance switches, betrayal, and volatility should amplify the precision channel. The desired result is not necessarily recovery success. A valid boundary-condition result is precision-driven misdeployment: lowered entropy, altered returns or reallocation, and worse outcomes after a sudden shift.

H4 tests social choice. Partner-specific precision should alter whom the agent selects, avoids, probes, or returns to, even when total payoff is still flat.

H5 tests whether clinical-like precision perturbations separate in beta dynamics and behavior. H5 is supplemental because current runs use five seeds per variant and because no human or clinical validation data are included.

4.2 Evidence base

The primary measures are total payoff, mean policy entropy, type/stance/joint belief accuracy, predictive log score, partner-selection entropy, partner-selection rates, beta statistics, precision-surprise and precision-payoff associations, reencounter behavior after betrayal, payoff conditional on return, and low-entropy misdeployment rates. Analysis outputs report mean differences, bootstrap confidence intervals for selected effects, and pairwise tests for payoff comparisons. This draft treats those statistics as current evidence but not as a final preregistered analysis package.

5 Results

5.1 Open policy spaces permit affective deployment

In the H0/H2 open graded regime, using five seeds per variant, affective precision changed behavior in the predicted direction. The affective agent reached total payoff 1884.6 with mean policy entropy 7.94, while the no-affect or lesioned baseline reached total payoff 1859.4 with mean policy entropy

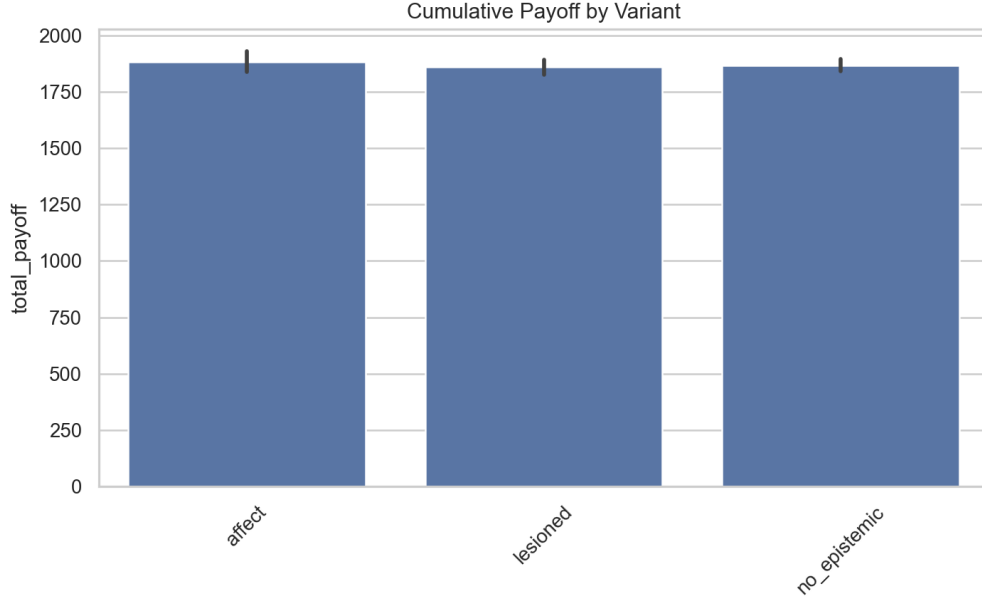


Figure 2: Open-regime deployment. In the open graded regime, affective precision lowers policy entropy and increases payoff relative to the baseline, while belief readouts remain similar or slightly favor the lesion. This diagnostic figure should be redrawn from source tables before submission.

8.80. The resulting difference was +25.2 payoff and -0.86 entropy for affect. Because this run uses five seeds per variant, it is supporting evidence rather than the manuscript’s strongest statistical anchor.

The interpretation is that open graded policy spaces allow partner-local precision to move the policy posterior. The result should not be generalized to all regimes. The same mechanism can be behaviorally invisible in saturated binary settings and harmful in abrupt betrayal settings. Thus, H0 supports a gate claim: policy openness is necessary for visible affective deployment, but it does not guarantee a payoff benefit.

5.2 Partner-specific precision tracks model fitness rather than reward

The H1 reliability-versus-reward confirmation is the cleanest mechanism result. With 30 seeds per variant and 200 rounds per seed, the affective agent did not outperform the no-affect baseline on payoff: total payoff was 534.6 for affect and 542.1 for no affect, a difference of -7.53 with bootstrap CI $[-27.88, 13.82]$. This absence of a payoff advantage is important because it prevents the result from being explained as a generic reward booster.

Despite no payoff advantage, the beta-derived precision signal tracked predictive reliability more strongly than reward. Across partner-seed units, $|\text{corr}(\text{precision}, \text{surprise})| = 0.701$, while $|\text{corr}(\text{precision}, \text{payoff})| = 0.419$. The reliability-over-reward effect was $+0.096$ with bootstrap CI $[0.027, 0.164]$. Mean policy entropy was 4.48 for affect and 4.32 for no affect, a difference of $+0.16$ with bootstrap CI $[0.06, 0.27]$. The main result is therefore a dissociation: partner-local precision follows predictive reliability more strongly than realized payoff.

This supports the model-fitness interpretation. If beta were merely caching reward, precision should

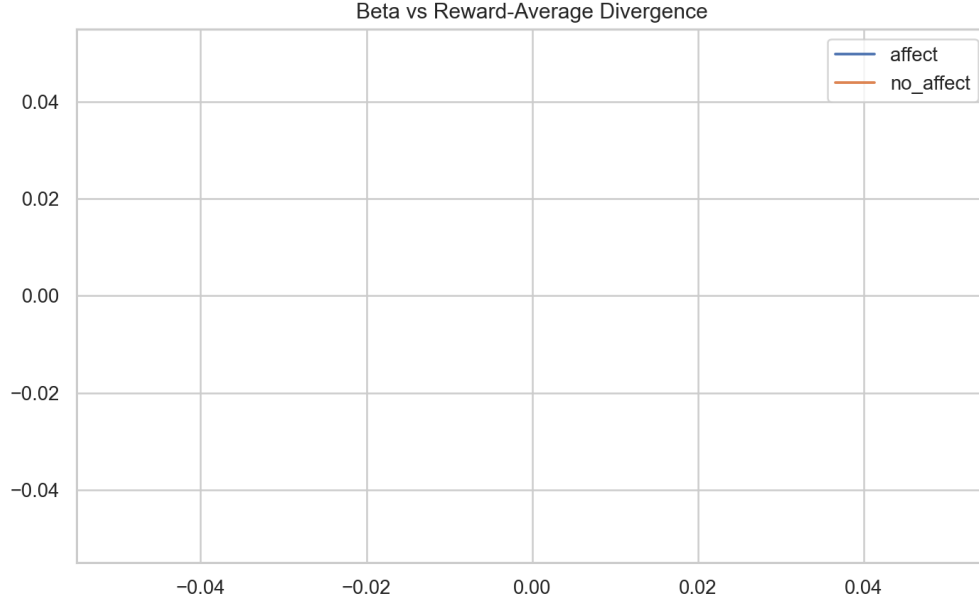


Figure 3: Model-fitness readout. In the 30-seed H1 confirmation, partner-local precision tracks predictive surprise more strongly than realized payoff, while affect has no total-payoff advantage.

collapse toward payoff history. Instead, it is more tightly associated with surprise and predictability. In social terms, the agent can be precise about a partner it should avoid. Precision is confidence in the social model, not affection for the partner.

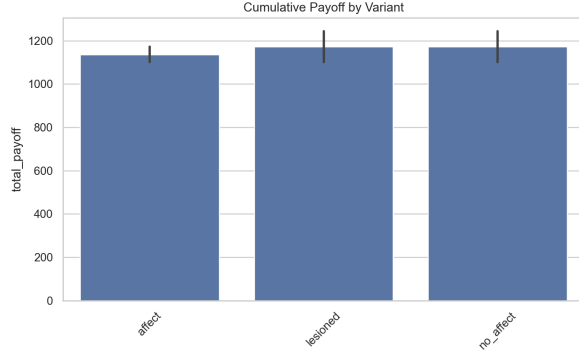
5.3 Affect changes deployment despite similar belief readouts

The H2 open-regime lesion tests whether beta affects action deployment rather than partner inference. With five seeds per variant, affect produced payoff 1884.6, joint accuracy 0.336, stance accuracy 0.816, and mean policy entropy 7.94. The tracked-only lesion produced payoff 1859.4, joint accuracy 0.339, stance accuracy 0.855, and mean policy entropy 8.80.

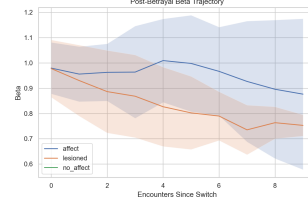
The belief readouts were therefore similar and in some cases slightly favored the lesion, while policy entropy and payoff favored the affective deployment pathway. This is the strongest current knowing-versus-using pattern: the lesion resembles an agent that can maintain partner-belief readouts but cannot use the partner-specific model-fitness signal to sharpen or soften policy selection. Because the run uses five seeds per variant, the manuscript treats it as supporting evidence that complements the primary 30-seed H1 and H3 results.

5.4 Partner choice moves before payoff

H4 tests whether partner-specific precision affects social selection. With five seeds per variant in the partner-choice run, payoff was nearly flat: affect reached 393.6 and no affect reached 393.2. The behavioral distribution nevertheless moved. Mean policy entropy was 3.99 for affect and 4.83 for no affect. Mean selection rates by partner were [0.366, 0.316, 0.138, 0.180] for affect and [0.348, 0.318, 0.162, 0.172] for no affect.



(a) Cumulative payoff



(b) Betrayal signal trajectories

Figure 4: Betrayal boundary condition. Abrupt betrayal makes partner-local precision behaviorally visible, but not safer: affect lowers entropy and changes reallocation while reducing whole-run payoff.

This result is useful because payoff is a downstream and sometimes insensitive readout in partner-choice settings. H4 predicts approach, avoidance, and probing shifts, not necessarily immediate payoff gain. The observed entropy and selection-rate changes show behavioral movement before total payoff becomes informative. As with H0 and H2, the five-seed sample means H4 should be presented as supporting behavioral evidence rather than a standalone statistical anchor.

5.5 Abrupt betrayal exposes a precision-driven boundary condition

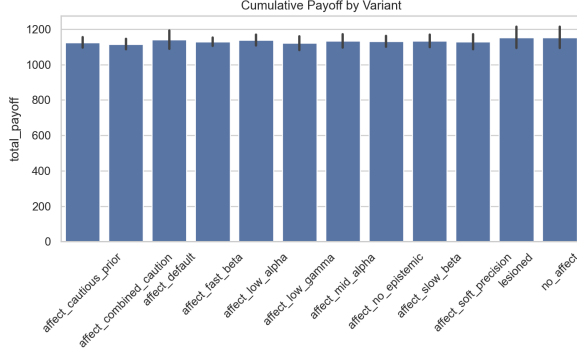
The H3 betrayal reallocation confirmation is the primary stress result. With 30 seeds per variant and 120 rounds per seed, affect reduced whole-run payoff relative to the no-affect or lesioned baseline. Total payoff was 1136.1 for affect and 1172.1 for no affect/lesion, a difference of -36.08 with bootstrap CI $[-63.65, -10.93]$ and pairwise $p = 0.0169$. Mean policy entropy was lower for affect: 8.38 versus 8.74, a difference of -0.36 with bootstrap CI $[-0.45, -0.26]$.

The post-switch reencounter readouts show an active but risky precision channel. Affect reencountered the switched partner 4.4 times on average, compared with 6.1 for no affect. Reencounter selection rate was 0.049 for affect and 0.067 for no affect. However, payoff conditional on reencounter did not improve: affect averaged 8.76 on reencounters, while no affect averaged 8.91. The wrong-type rate on reencounter was 0.237 for affect and 0.165 for no affect.

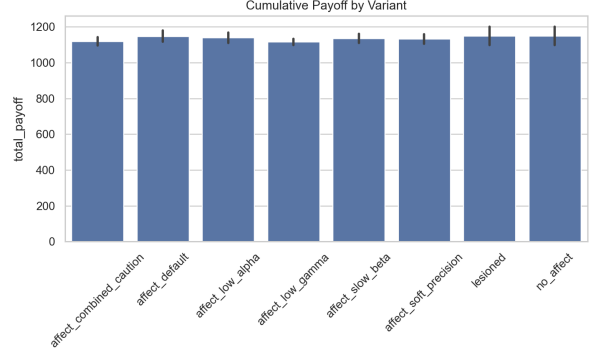
The correct interpretation is not that affect recovers after betrayal. The correct interpretation is that abrupt betrayal reveals the precision channel and its failure mode. Affect lowers entropy and changes return/reallocation behavior, but it does not improve return safety or whole-run payoff in the 30-seed confirmation. Under abrupt social shocks, precision can sharpen action around a model that has not recalibrated fast enough.

5.6 Shock shape matters more than generic caution

The H3 precision-sensitivity follow-up tests whether the betrayal failure is simply excessive gain that can be fixed by generic caution. In the abrupt sensitivity run, with 30 seeds per variant and 120 rounds per seed, the no-affect/lesion baseline had payoff 1153.6 and entropy 8.80. Default affect



(a) Abrupt betrayal



(b) Gradual betrayal

Figure 5: Shock-shape sensitivity. Default affect is costly under abrupt betrayal but nearly payoff-neutral under gradual betrayal. Cautious variants raise entropy but do not rescue payoff.

had payoff 1140.4 and entropy 8.35, with pairwise $p = 0.370$ against no affect. Combined caution had payoff 1115.0 and entropy 9.27, with $p = 0.003$ against no affect.

The gradual sensitivity run changed the shock shape. With 30 seeds per variant, the no-affect/lesion baseline had payoff 1148.9 and entropy 8.76. Default affect was nearly payoff-neutral, with payoff 1147.6, entropy 8.38, and $p = 0.906$ against no affect. Combined caution again reduced payoff, reaching 1118.3 with entropy 9.25 and $p = 0.005$ against no affect.

These results argue against a simple “make the agent more cautious” repair. Generic caution can raise entropy but still harm payoff. The more informative axis is shock shape: default affect is costly mainly under abrupt betrayal, while gradual betrayal largely removes the default affect payoff penalty. The boundary condition appears to be temporal credit-assignment mismatch under sudden model change, not merely too much policy precision.

5.7 Clinical-like perturbations separate precision dynamics, not clinical validity

H5 is supplemental. In the five-seed dynamics run, total payoff was 1884.6 for affect, 1871.1 for borderline-like, 1840.7 for alexithymia-like, and 1834.1 for depression-like variants. The more important readout is beta dynamics. Beta range was 1.318 for borderline-like, 1.435 for depression-like, 0.950 for default affect, and 0.149 for alexithymia-like. In the five-seed betrayal run, payoff was 1216.3 for alexithymia-like, 1196.9 for borderline-like, 1168.3 for affect, and 1150.1 for depression-like.

These numbers show that parameter perturbations can separate in precision range, entropy, and selection dynamics. They do not validate clinical diagnoses. The current runs are underpowered for payoff claims and task-specific. H5 belongs in the supplement unless the paper is reframed around perturbation dynamics.

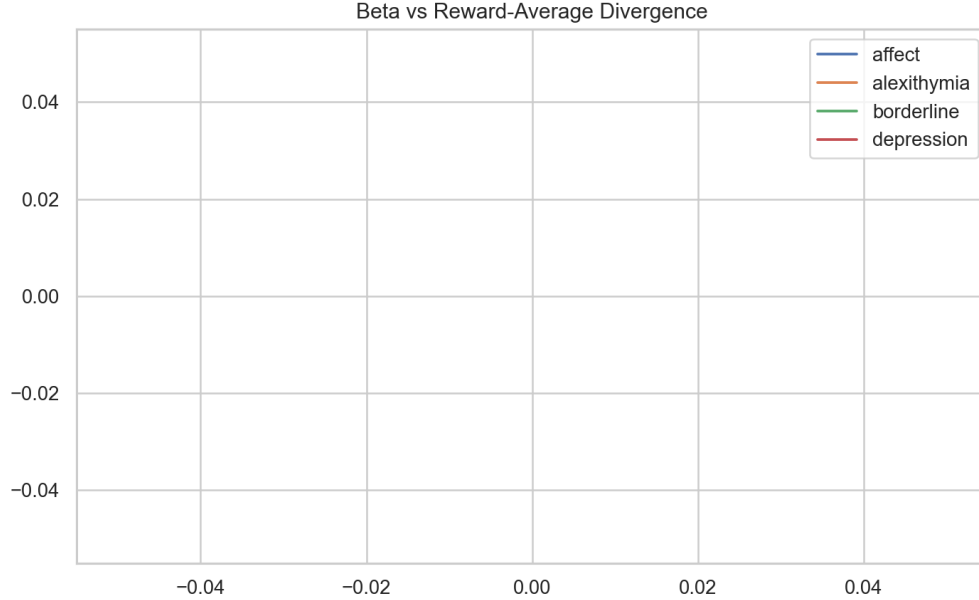


Figure 6: Clinical-like perturbation dynamics. Perturbations separate in beta dynamics and policy/choice readouts, but these are computational parameter variants rather than validated clinical phenotypes.

6 Discussion

6.1 Main interpretation

The current evidence supports partner-local affective precision as computational infrastructure for social policy deployment. H1 shows that precision is more closely tied to predictive reliability than realized payoff. H2 shows that decoupling beta from policy precision can preserve similar belief readouts while changing entropy and payoff. H4 shows that partner-choice distributions can move before payoff moves. H3 shows that the same precision channel can become harmful when social volatility is abrupt.

This combination is stronger than a monotonic reward-improvement story. A generic reward booster would predict payoff gains and would not naturally explain why precision follows surprise more strongly than reward. A pure belief-accuracy story would predict improvements in type or stance readouts, but H2 shows that belief readouts can remain similar while deployment changes. A pure caution story would predict that making the agent more cautious repairs betrayal failure, but the H3 sensitivity results show that generic caution can increase entropy while reducing payoff.

The model therefore occupies a middle position. Partner-local affective precision is useful because it compresses model fitness by partner and gates action on the basis of that confidence. It is dangerous because confidence can be misplaced during abrupt model mismatch. Affect is not a guarantee of safer recovery; it is a mechanism that can improve or impair deployment depending on whether the underlying partner model is calibrated to the current regime.

6.2 Why this is not reward, trust, or deeper planning

The H1 dissociation is the central argument against reward averaging. In the 30-seed H1 confirmation, affect had lower total payoff than no affect by 7.53 points, with a confidence interval spanning zero, yet precision tracked surprise more strongly than payoff. This is exactly what a model-fitness account predicts. The affective state says “my model of this partner is reliable” rather than “this partner is good for me.”

The mechanism is also not a literal trust scalar. Trust can mean willingness to rely on a partner, expected benevolence, or learning-rate modulation. Here, beta modulates policy precision. A predictable adversary can have high model precision but low approach value. A future model could add a separate trust-as-learning-rate variable, but the current implementation intentionally keeps trust and affective precision distinct.

The mechanism is not simply deeper planning. The current architecture uses action-dependent stance dynamics so planning depth is meaningful, but the manuscript’s claims concern partner-local precision as an orthogonal deployment signal. Policy precision can change how the agent uses a policy set; it is not another step of lookahead. This is why policy openness matters: if the policy posterior is already saturated, precision changes may be invisible. If several policies remain plausible, precision can strongly alter behavior.

6.3 Boundary conditions as evidence

The H3 betrayal result should be foregrounded rather than hidden. It shows that the mechanism is active under stress but not necessarily beneficial. Abrupt betrayal creates a temporal mismatch: the beta channel can sharpen policy selection before the partner model has recalibrated. Lower entropy is then not safer action; it is confident action under a wrong or incomplete model.

This boundary condition is theoretically informative. It predicts that affective deployment should be most helpful when reliability estimates track regime changes at the right timescale, and most harmful when social structure changes faster than the model can update. The gradual-betrayal sensitivity result is consistent with this interpretation: default affect becomes nearly payoff-neutral when the shock is less abrupt.

6.4 Literature positioning

The project extends Hesp-style affective precision from a single-agent expected-action-precision account to partner-specific social model-fitness estimates [TODO-CITE: Hesp et al. *Deeply Felt Affect*]. It is complementary to multi-agent active-inference work that explicitly models other agents for strategic planning [TODO-CITE: *factorised active inference for strategic multi-agent interactions*]. Its contribution is not a general multi-agent planning algorithm; it is the partner-local affective precision signal and the behavioral tests that separate model fitness from reward.

The trust task also sits near active-inference models of trust and automation [TODO-CITE: *active inference trust automation papers*]. The manuscript should not claim to be the first active-inference model of trust. The narrower novelty claim is stronger: the model separates partner-specific affective precision from reward and from a dedicated trust scalar, then tests it with model-fitness, deployment-lesion, social-choice, and betrayal-stress readouts.

The H5 variants should be positioned carefully relative to computational psychiatry and active-inference psychiatry reviews [TODO-CITE: active inference psychology and psychiatry reviews]. They are not clinical validation. They are parameter perturbations showing that beta dynamics can separate in ways that may be useful for future computational phenotype hypotheses.

7 Limitations

First, the evidence is simulation-only. It supports a mechanism claim inside a trust-task testbed, not a claim about human affect, human trust, or clinical diagnosis. Human behavioral fitting or experimental data would be needed before claiming psychological validity.

Second, the primary trust task uses environment-owned partner behavior. This is appropriate for isolating the focal agent’s partner-local precision mechanism, but it does not establish fully reciprocal multi-agent dynamics. AIF-vs-AIF partner interactions remain future or descriptive work until promoted into specific hypotheses.

Third, partner-specific beta has not yet been compared against a global-beta ablation. Partner specificity is an architectural premise of the current model. A global-beta control would directly test whether a shared affective precision state can explain the same partner-choice, entropy, payoff, and model-fitness results.

Fourth, H0, H2, H4, and H5 use five seeds per variant. They are useful supporting evidence but should not carry the manuscript’s strongest claims. H1 and H3, each with 30 seeds per variant, should remain the empirical anchors.

Fifth, H3 is a boundary condition rather than a recovery win. Abrupt betrayal reduces affective-agent payoff in the 30-seed confirmation. The paper should state this plainly. The mechanism changes deployment and reallocation but does not produce safer return behavior under that shock regime.

Sixth, H5 payoff claims are underpowered and task-dependent. The clinical-like variants separate in precision dynamics, but diagnostic language should be avoided. The correct terms are “clinical-like parameter perturbations” or “computational perturbation variants.”

Seventh, the current figures are draft analysis aids. Camera-ready figures should be redrawn as multi-panel composites from source tables, with consistent scales, uncertainty intervals, and captions that state seed counts.

8 Future Work

The highest-value follow-up is a global-beta ablation. The model’s architectural novelty is partner-specific affective precision, but the current evidence does not directly compare partner-local beta against one shared beta tracker. A global-beta variant should share one beta posterior across partners while keeping the same $\gamma = \gamma_{\text{base}}/\mathbb{E}[\beta]$ mapping. The most useful tests would be H1 reliability-versus-reward, H2 open-regime lesion, and H4 partner choice, with optional H3 stress checks.

A second useful follow-up is a focused shock-shape gradient. The current H3 sensitivity results suggest that abruptness, not generic overconfidence, drives the betrayal failure. A focused experi-

ment could hold default affect fixed while varying the abruptness of the stance switch, observation noise, or number of pre-switch confirmations. The primary readouts should be payoff, entropy, reencounters, payoff-on-return, wrong-type rate, and low-entropy wrong-rate.

A third near-term need is figure-quality composite generation. Current PNGs are diagnostic aids. A script that reads the copied source tables and emits final multi-panel PNG/PDF figures would likely improve the manuscript more than another broad sweep.

Longer-term work includes human-data validation, model comparison against reward-averaging and belief-only baselines, a fuller variational beta implementation, richer social environments with correlated partners, and reciprocal AIF agents interacting with each other.

9 Conclusion

Partner-specific affective precision is a plausible computational bridge between social model fitness and social policy deployment. In the current simulations, it is behaviorally visible when policy spaces are open, tracks predictive reliability more than payoff, changes action deployment despite similar belief readouts, and alters partner choice before payoff clearly moves. Under abrupt betrayal, the same mechanism can sharpen the wrong deployment state and reduce payoff. The current evidence therefore supports a narrow but useful claim: affective precision is a partner-local model-fitness signal, not a generic reward booster. Future work should test the partner-specific architecture against global beta, validate the mechanism against human behavior, and map the shock-regime boundary conditions more precisely.

References to verify before submission

- Hesp et al., “Deeply Felt Affect: The Emergence of Valence in Deep Active Inference.” [TODO-CITE: verify exact bibliographic details]
- Friston and related active-inference / expected-free-energy sources. [TODO-CITE: verify exact bibliographic details]
- Parr, Pezzulo, and Friston active-inference textbook or review. [TODO-CITE: verify exact bibliographic details]
- Damasio somatic-marker hypothesis and Bechara Iowa Gambling Task papers. [TODO-CITE: verify exact bibliographic details]
- Recent reviews on active inference in psychology and psychiatry. [TODO-CITE: verify exact bibliographic details]
- Factorised active inference for strategic multi-agent interactions. [TODO-CITE: verify exact bibliographic details]
- Active-inference models of trust or human-automation trust. [TODO-CITE: verify exact bibliographic details]

A Claim and Non-Claim Checklist

Table 3: Draft claim discipline.

Claim	Current support	How to write it
Policy openness gates affect effects	H0/H2 five-seed open-regime evidence; saturated regimes weaker	“Policy openness gates visible affective deployment.”
Partner-local precision tracks model fitness more than reward	H1 30-seed confirmation; precision-surprise association exceeds precision-payoff association	“Precision is predictive reliability, not cached partner value.”
Affect changes deployment more than belief inference	H2 five-seed lesion; belief readouts similar while entropy/payoff move	“Knowing-without-using dissociation.”
Stress amplifies the channel	H3 30-seed betrayal evidence; entropy lower and behavior changes	“Stress exposes both mechanism and failure mode.”
Affect improves recovery after betrayal	Not supported in current H3 confirmation	Do not claim. Say abrupt betrayal exposes misdeployment risk.
Generic caution fixes H3	Not supported; cautious variants often hurt payoff	Say shock shape matters more than generic caution.
Partner-specific precision guides social choice	H4 five-seed partner-choice shift with flat payoff	“Partner choice is the readout; payoff is secondary here.”
Clinical variants are validated phenotypes	Not supported	Say “clinical-like parameter perturbations,” not diagnoses.
Partner-local beta beats global beta	Not yet tested	List as future global-beta ablation.

B Figure Plan

1. **Figure 1: Model schematic.** Show partner action prediction error $\rightarrow$ partner-specific β_k / precision $_k \rightarrow$ partner-local $\gamma_k \rightarrow$ policy posterior shift $\rightarrow$ action/partner choice $\rightarrow$ payoff, reallocation, or misdeployment. Include official partner-local `pymdp.Agent` instances, hidden factors, observations, external beta tracker, beta-to-gamma modulation, and tracked-only lesion.
2. **Figure 2: Open-regime deployment.** Show total payoff, policy entropy, and belief readouts for affect versus no-affect/lesion in the open graded regime.
3. **Figure 3: Model fitness versus reward.** Show precision-surprise and precision-payoff associations, reliability-over-reward effect, and no payoff advantage.
4. **Figure 4: Betrayal boundary condition.** Show payoff, entropy, reenounters, payoff-on-return, and wrong-type rate.

5. **Figure 5: Shock shape and sensitivity.** Compare abrupt and gradual betrayal, including default affect, no-affect/lesion baseline, and cautious variants.
6. **Supplement: H4 social choice and H5 perturbation dynamics.** Show partner-selection rates, partner-selection entropy, beta range, beta autocorrelation, action-flip rate, and selection entropy.

C Result Summary Table

Table 4: Numerical result summary for draft writing.

Result	Primary numbers	Interpretation
R1 H0/H2 openness	Five seeds: affect payoff 1884.6, entropy 7.94; baseline payoff 1859.4, entropy 8.80	Open graded policy spaces let precision move deployment.
R2 H1 model fitness	30 seeds: payoff 534.6 vs 542.1; $ r_{\text{precision,surprise}} = 0.701$ vs $ r_{\text{precision,payoff}} = 0.419$; reliability-over-reward +0.096 CI [0.027, 0.164]	Precision tracks predictive reliability more than reward.
R3 H2 lesion	Five seeds: affect joint accuracy 0.336, stance accuracy 0.816, entropy 7.94; lesion joint accuracy 0.339, stance accuracy 0.855, entropy 8.80	Beliefs similar; deployment differs.
R4 H4 social choice	Five seeds: payoff 393.6 vs 393.2; entropy 3.99 vs 4.83; selection rates move from [0.348, 0.318, 0.162, 0.172] to [0.366, 0.316, 0.138, 0.180]	Choice distribution changes before payoff.
R5 H3 betrayal	30 seeds: payoff 1136.1 vs 1172.1; CI $[-63.65, -10.93]$; entropy 8.38 vs 8.74; reencounters 4.4 vs 6.1; wrong-type rate 0.237 vs 0.165	Abrupt betrayal exposes misdeployment.
R6 H3 shock shape	30 seeds: abrupt baseline 1153.6, default affect 1140.4, combined caution 1115.0; gradual baseline 1148.9, default affect 1147.6, combined caution 1118.3	Gradual shock removes default affect payoff penalty; generic caution does not rescue payoff.
R7 H5 perturbations	Five seeds: dynamics beta range borderline 1.318, depression 1.435, affect 0.950, alexithymia 0.149	Perturbations separate dynamics, not clinical validity.